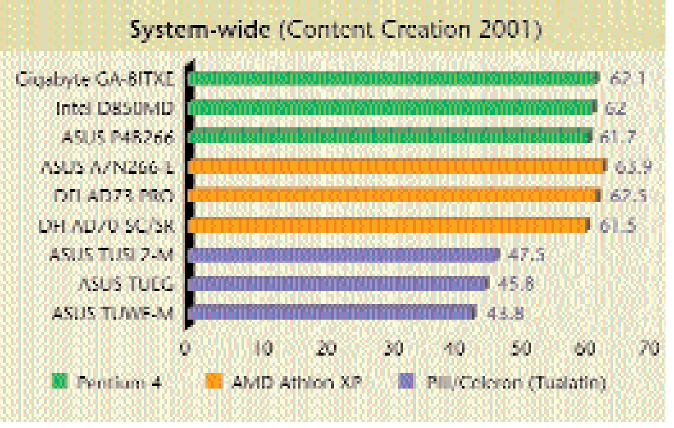

P4B266 which scored 61.7 in CCWS and finished the video encoding sequence in 116 seconds. This performance was matched by both the MSI 845 Ultra-C and the MSI 845 Ultra-ARU. Though they had lower scores in CCWS (59.1 and 60.4 respectively) they were still ahead of the fastest SDRAM board—the Mercury KOB845NFSX which scored 58.1 in CCWS. This board took an entire 131 seconds to complete the video encoding sequence!

Athlon XP/Duron: The nForce 420-D-based ASUS A7N266-E (using DDR memory) beat the fastest Pentium 4 solution in the CCWS test with an amazing score of 63.9 and then finished the video encoding test in 106 seconds flat. The DFI AD73Pro came in second scoring 62.5 in CCWS and took around 110 seconds to finish encoding the test video.

Pentium III/Celeron (Tualatin): As we've observed in the previous tests, the ASUS TUSL2-M ends up dominating this category once again by scoring 47.5 in CCWS and then finishing the video encoding sequence in 154 seconds. All the other SDRAM boards trailed far behind.

Despite using the same processor and sometimes even the same chipsets, the scores obtained with motherboards from different manufacturers varied considerably. While this might seem confusing, the reason is actually pretty simple. This difference in performance can be attributed to tweaking (or slightly bending the rules) on the part of certain manufacturers who increase the FSB frequency beyond the prescribed 100/133 MHz. Doing this to a small degree (up to 101.6 MHz or 134.6 MHz) is perfectly fine if the board uses higher quality components with greater tolerances. However, it's akin to overclocking and the manufacturers who choose to do this gain a performance edge over the competition.

Features that matter!

While two motherboards might appear similar in features offered on the surface, digging deeper brings a lot of differences to light. You need to remember that these features have a direct or indirect impact on overall system performance. All motherboards now come with a standard set of features such as two USB 1.1 ports, two DIMM slots, ATA-100 support and the like.

There are a few motherboards that incorporate cutting-edge features that set them apart from the competition. The MSI KT3 Ultra-ARU board uses the KT333 chipset and has native support for DDR333 memory and ATA-133 drives. It features an onboard IDE RAID controller (Promise 2076), which supports both RAID 0 and RAID 1. The packaging and presentation of this motherboard is also outstanding. With a total of eight USB ports, four for USB 2.0 and four for USB 1.1, along with a separate SPDIF S-bracket and a detailed manual, there is little that MSI does wrong. Add to this its plethora of overclocking options available in the BIOS and you have a motherboard that is truly built for the enthusiast!

Not to be left far behind are the ASUS P4B266 and the MSI 845 Ultra-ARU. These were the most feature rich Pentium 4 boards and incorporated some really nifty features such as integrated six-channel audio. MSI's ARU (Audio/RAID/USB) versions of its motherboards seem to have every single feature that you'll ever need.

But in the end, the motherboard that truly takes the cake is the ASUS A7V333. Along with goodies like USB 2.0, RAID suport, six-channel audio and ATA-133 support, it also comes with an IEEE 1394

Big value! Integrated Chipset

Not everyone needs a GeForce3 graphics card along with a 1 GHz+ processor. Motherboards with integrated chipsets, though considered low performing, satisfy a very important bracket of the computing pie—pure productivity. These are the Maruti 800s of the motherboard world; they run well, support most of the major features and, more importantly, perform decently.

This market was earlier dominated by SiS, ALi and VIA to a certain extent. The SiS 630 series has been quite popular but the current favourite is the 810 series for the Intel Pentium III processors.

Motherboards built on this rather tried and tested chipset are still being manufactured and now support the latest Tualatin processors in their 810E2 *avatar*. We received a fair number of boards based on this revised chipset with some of them performing well in the office productivity benchmarks.

The ASUS TUWE-M is based on this chipset and its performance is noteworthy in the graphics and disk subsystem tests where it outshone the SiS 630ET chipset-based ASUS TUSI-M and the 810-based Mercury KOB 810E TFSX. Since there is no AGP slot on these motherboards, all benchmarks were run using the onboard VGA with the default BIOS settings. The performance drop in the graphics department is tremendous but if you are not interested in gaming and are looking for pure functionality, then these boards fit like a glove.

There are a fair number of boards available for the AMD platform too, like the VIA ProSavage KM133A and VIA KLE133. These boards have the Savage4 AGP integrated into the board and provide pretty decent graphics performance. Another fine example is the Krypton M7VKS (using the KM133A), which outperforms the DFI AM33-EC, Gigabyte GA-7VMM and Soltek SL-75LIV, all of which are based on the VIA KLE133 integrated chipset.

None of the Pentium 4 motherboards that we received had integrated audio and video.